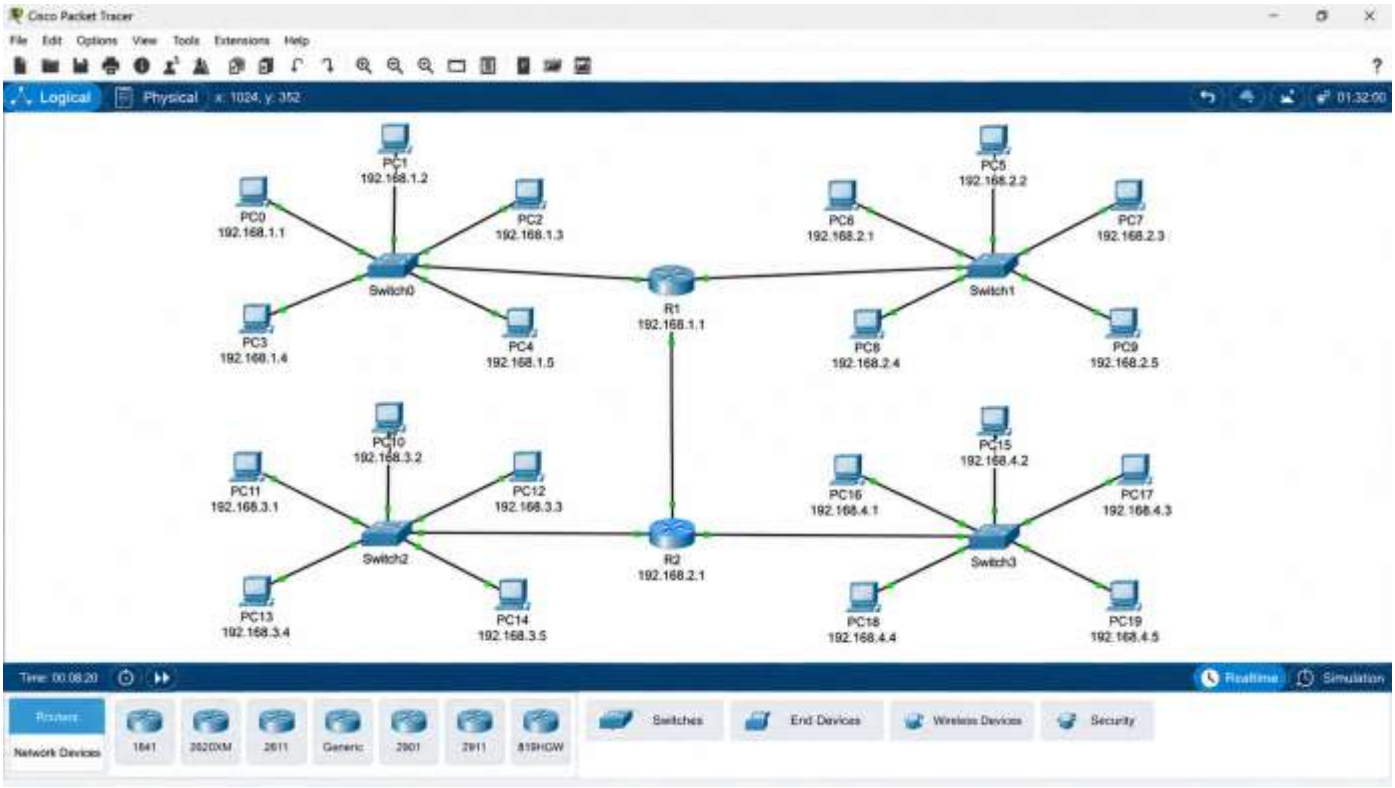
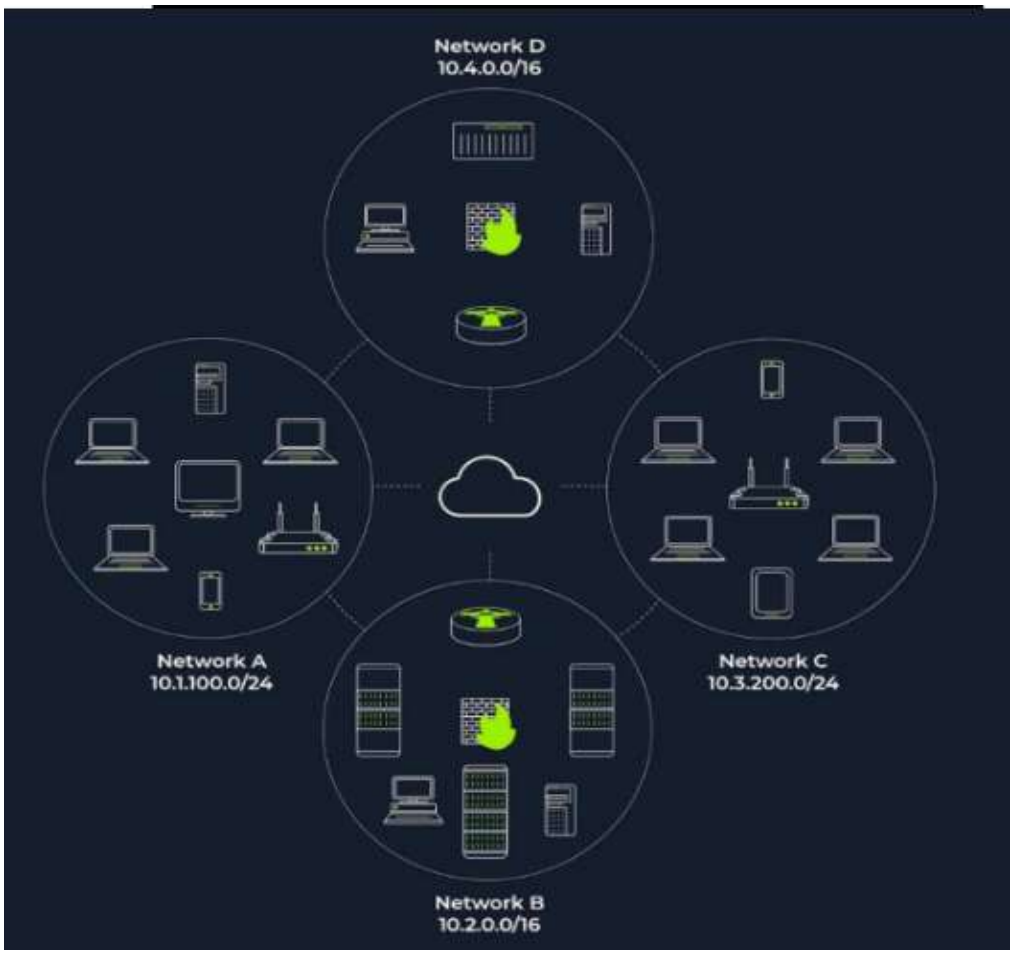

Cisco Packet Tracer
File Edit Options View Tools Extensions Help
Logical
Physical
PC0
192.168.1.1
PC1
192.168.1.2
PC2
192.168.1.3
PC3
192.168.1.4
PC4
192.168.1.5
Switch0
R1
192.168.1.1
PC5
192.168.2.2
PC6
192.168.2.1
PC7
192.168.2.3
PC8
192.168.2.4
PC9
192.168.2.5
Switch1
PC10
192.168.3.2
PC11
192.168.3.1
PC12
192.168.3.3
PC13
192.168.3.4
PC14
192.168.3.5
Switch2
R2
192.168.2.1
PC15
192.168.4.2
PC16
192.168.4.1
PC17
192.168.4.3
PC18
192.168.4.4
PC19
192.168.4.5
Switch3
Realtime
Simulation
Routers
Network Devices
Switches
End Devices
Wireless Devices
Security


Network D
10.4.0.0/16
Network A
10.1.100.0/24
Network C
10.3.200.0/24
Network B
10.2.0.0/16

Send Packet

From:

computer1

To:

computer1

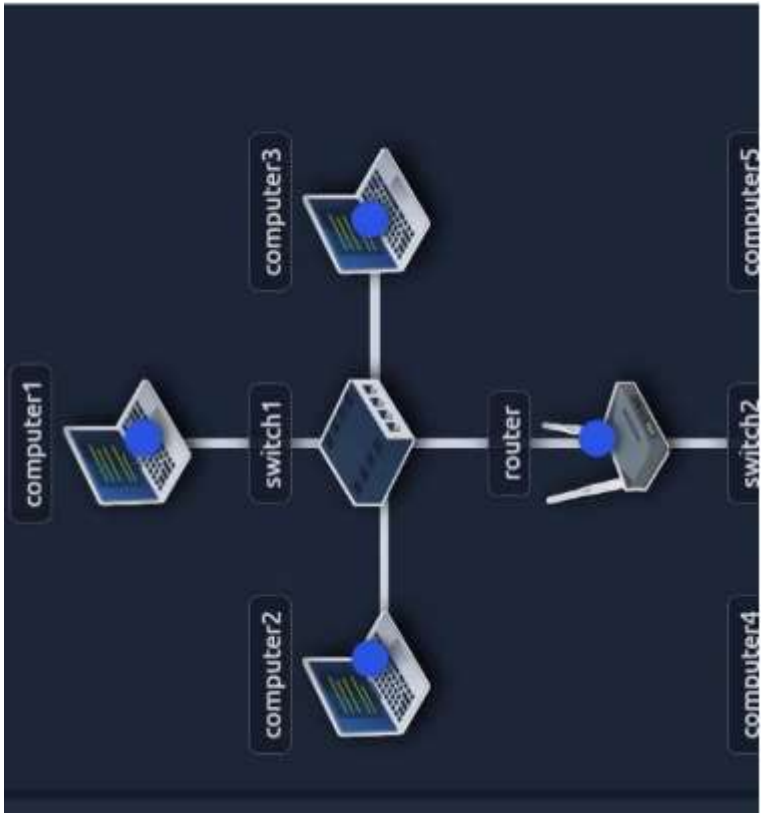
Packet Type:

arp_request

Data:

computer6

Send Packet



Send Packet

From:

computer4

To:

computer4

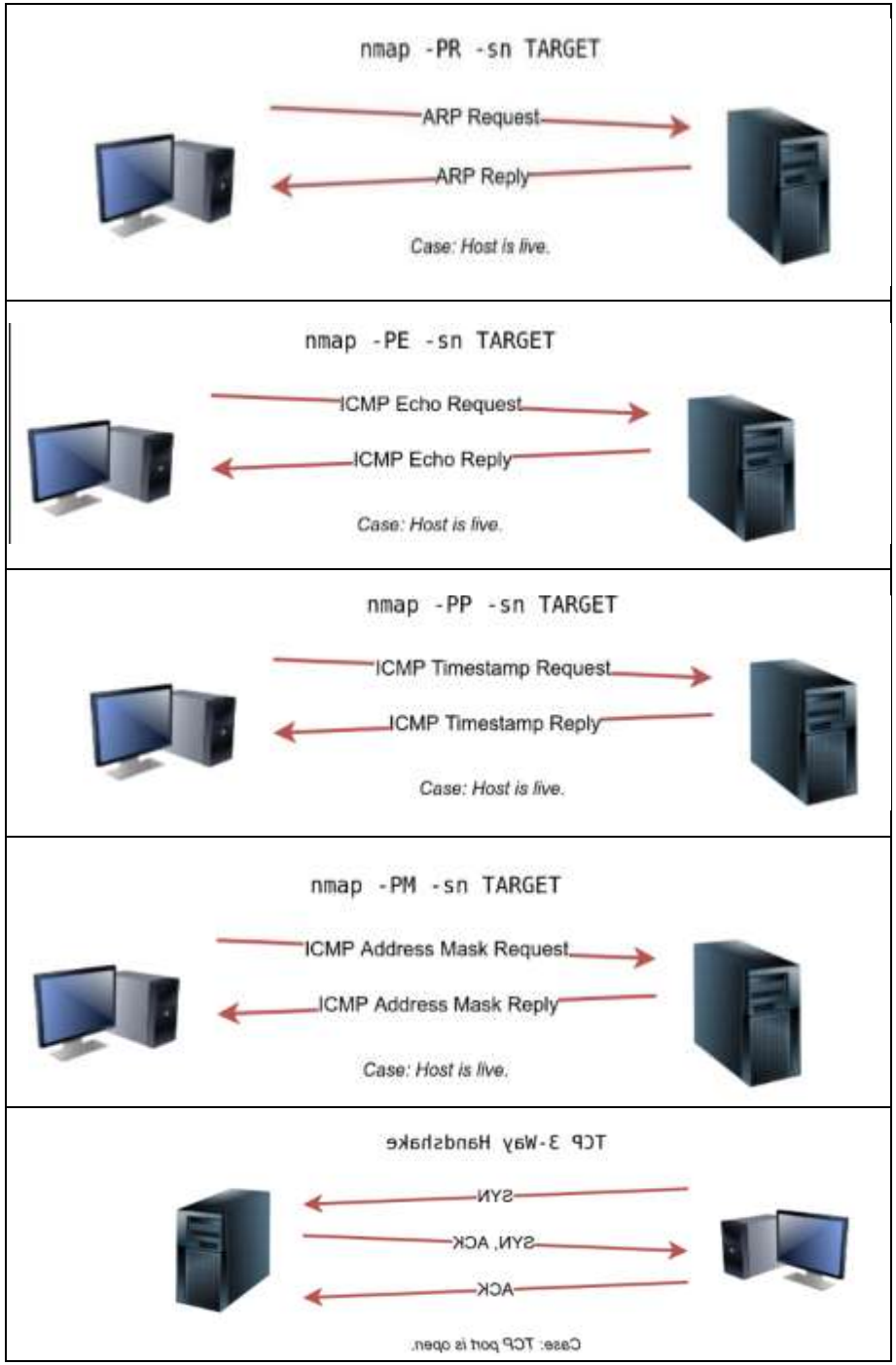
Packet Type:

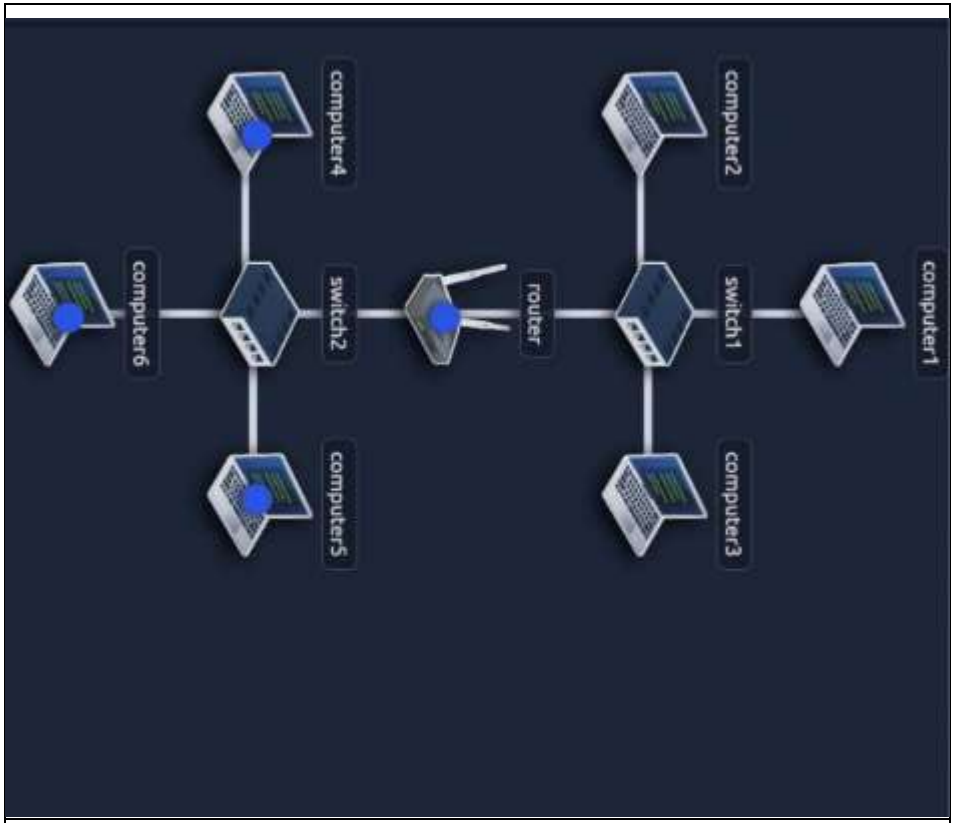
arp_request

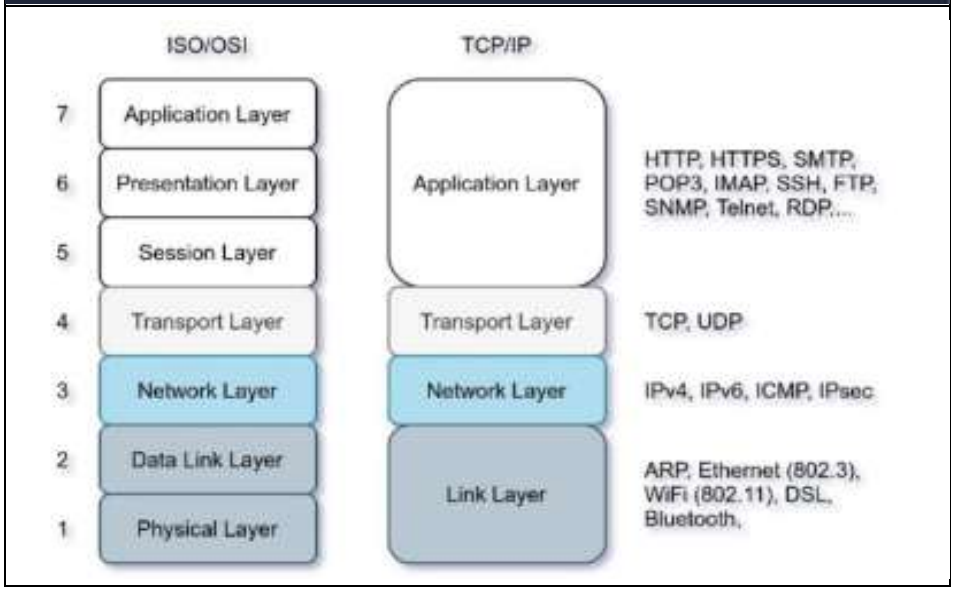
Data:

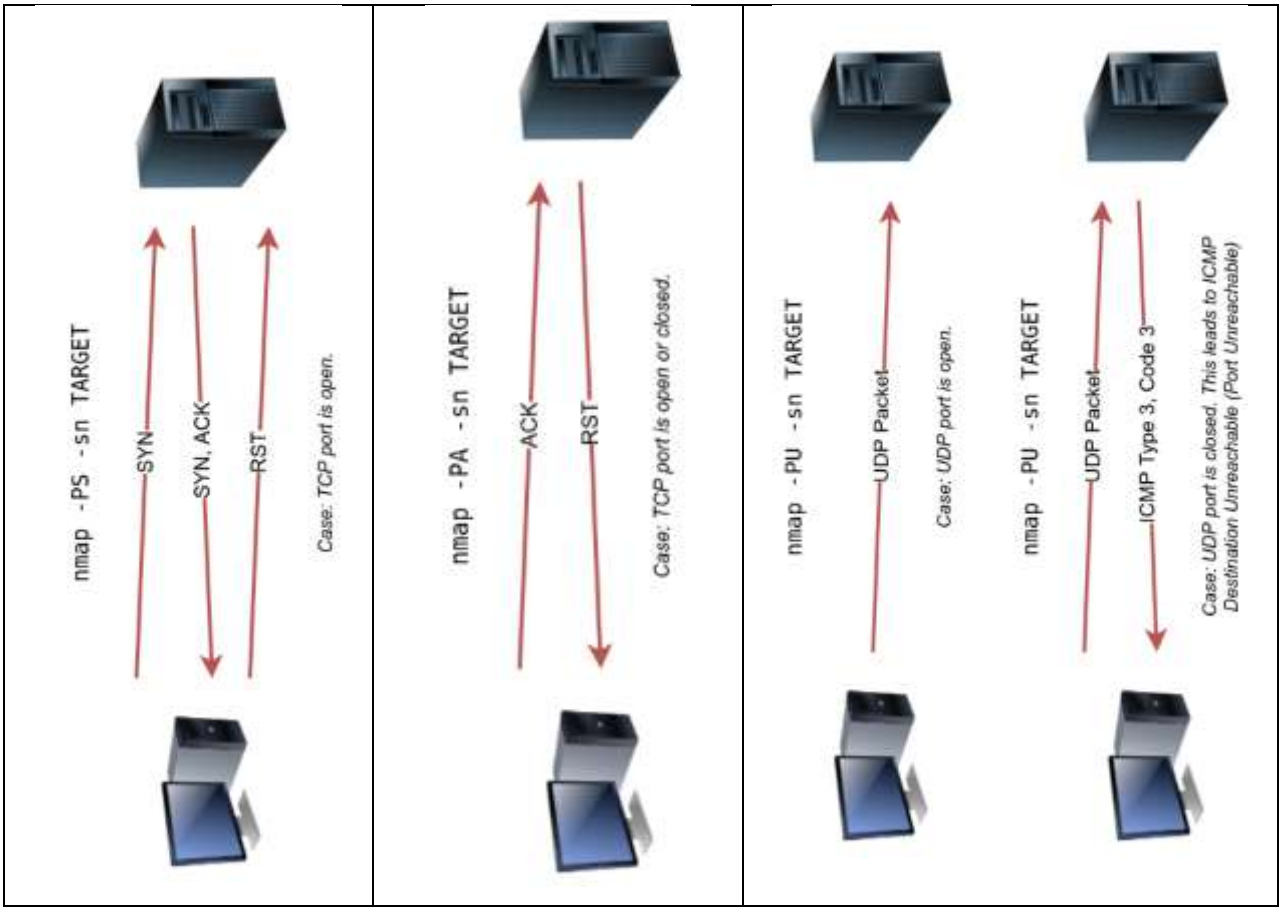
computer6

Send Packet


nmap -PR -sn TARGET
ARP Request
ARP Reply
Case: Host is live.
nmap -PE -sn TARGET
ICMP Echo Request
ICMP Echo Reply
Case: Host is live.
nmap -PP -sn TARGET
ICMP Timestamp Request
ICMP Timestamp Reply
Case: Host is live.
nmap -PM -sn TARGET
ICMP Address Mask Request
ICMP Address Mask Reply
Case: Host is live.

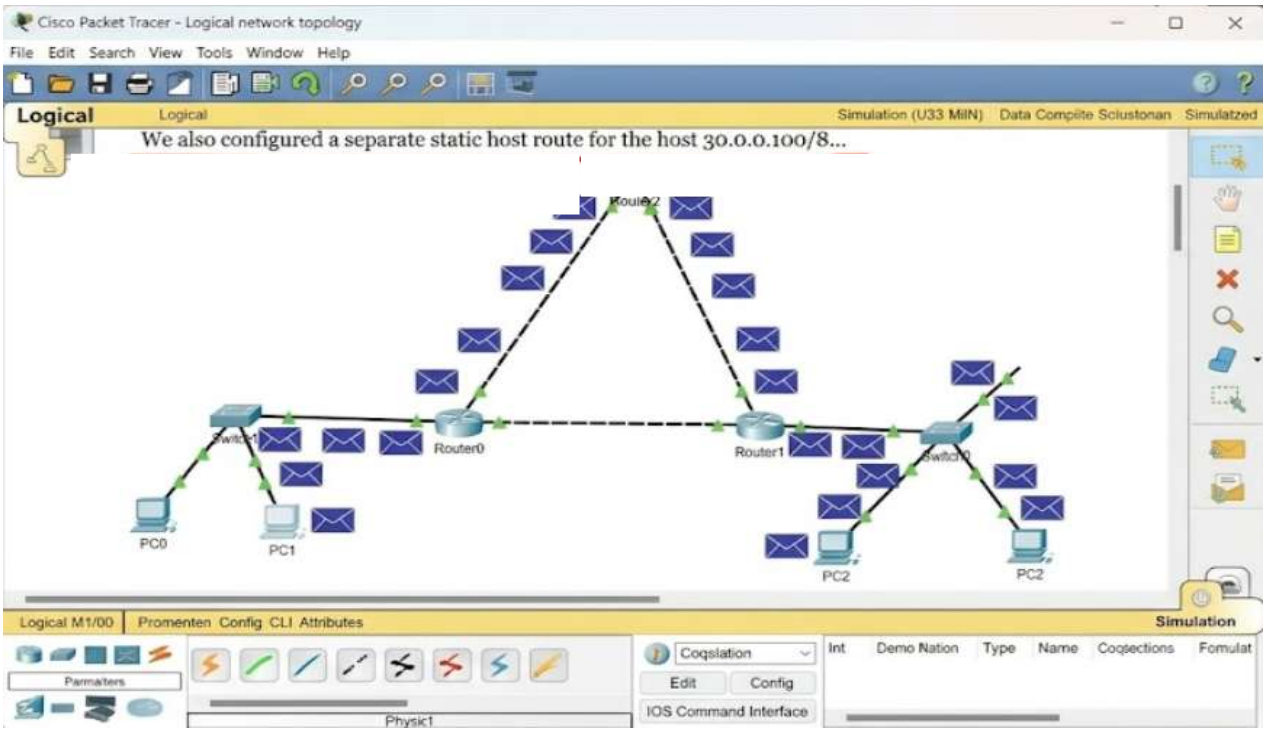

computer4
computer2
computer6
switch2
router
switch1
computer1
computer5
computer3

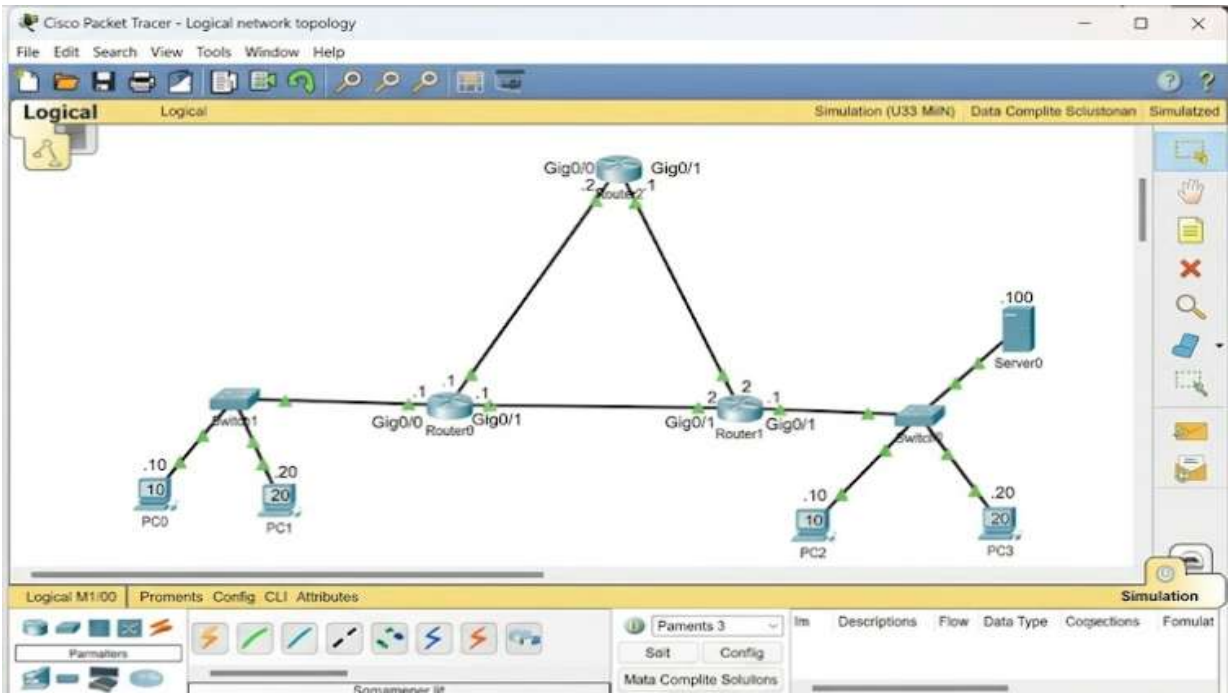

ISO/OSI
TCP/IP
7 Application Layer
6 Presentation Layer
5 Session Layer
4 Transport Layer
3 Network Layer
2 Data Link Layer
1 Physical Layer
Application Layer
Transport Layer
Network Layer
Link Layer
HTTP, HTTPS, SMTP, POP3, IMAP, SSH, FTP, SNMP, Telnet, RDP,...
TCP, UDP
IPv4, IPv6, ICMP, IPsec
ARP, Ethernet (802.3), WiFi (802.11), DSL, Bluetooth,

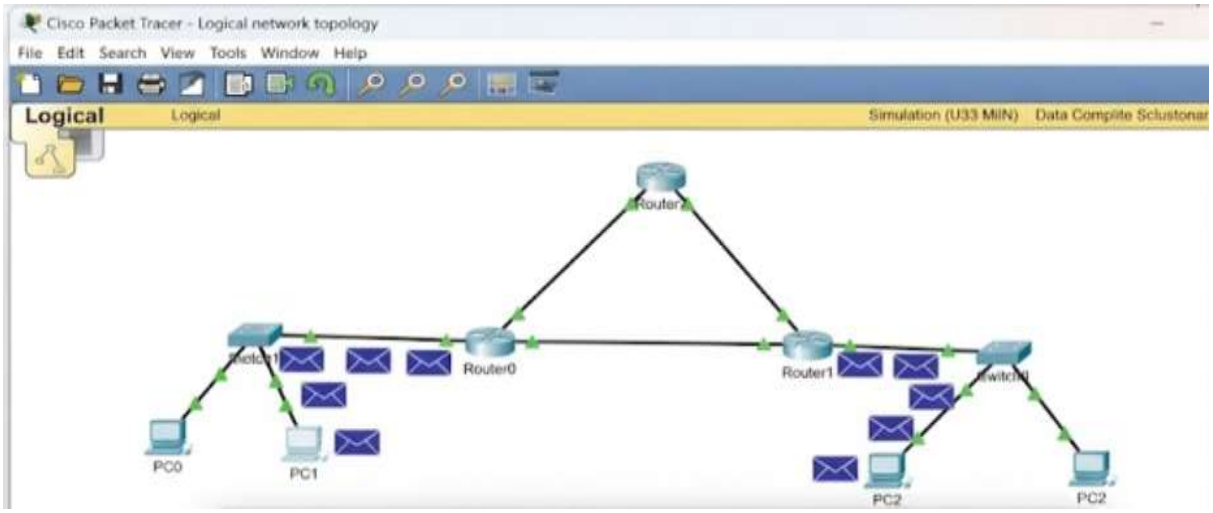

nmap -PS -sn TARGET
SYN
SYN, ACK
RST
Case: TCP port is open.
nmap -PA -sn TARGET
ACK
RST
Case: TCP port is open or closed.
nmap -PU -sn TARGET
UDP Packet
Case: UDP port is open.
nmap -PU -sn TARGET
UDP Packet
ICMP Type 3, Code 3
Case: UDP port is closed. This leads to ICMP Destination Unreachable (Port Unreachable)

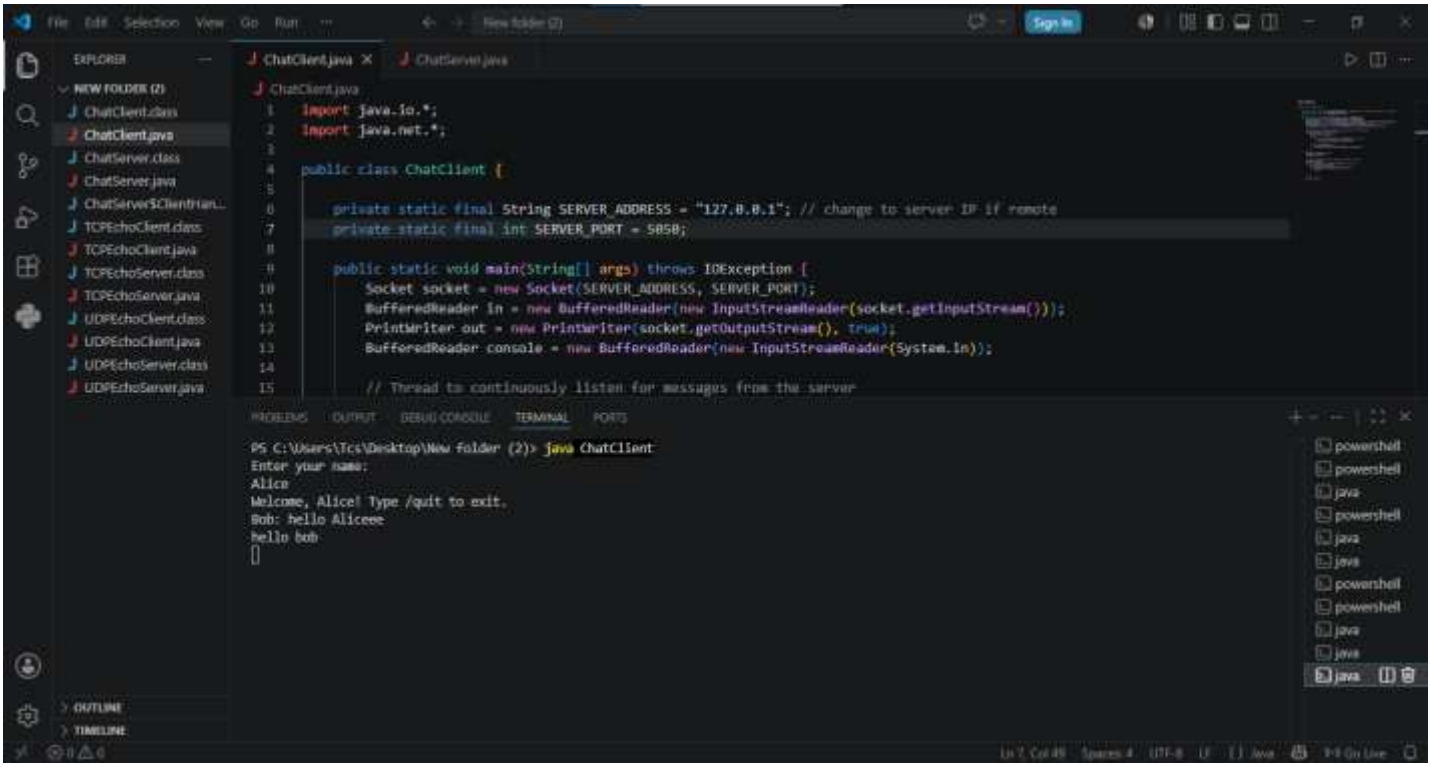

PC-PT PC0
PC-PT PC1
PC-PT PC2
Linksys-WRT300N Wireless Router0

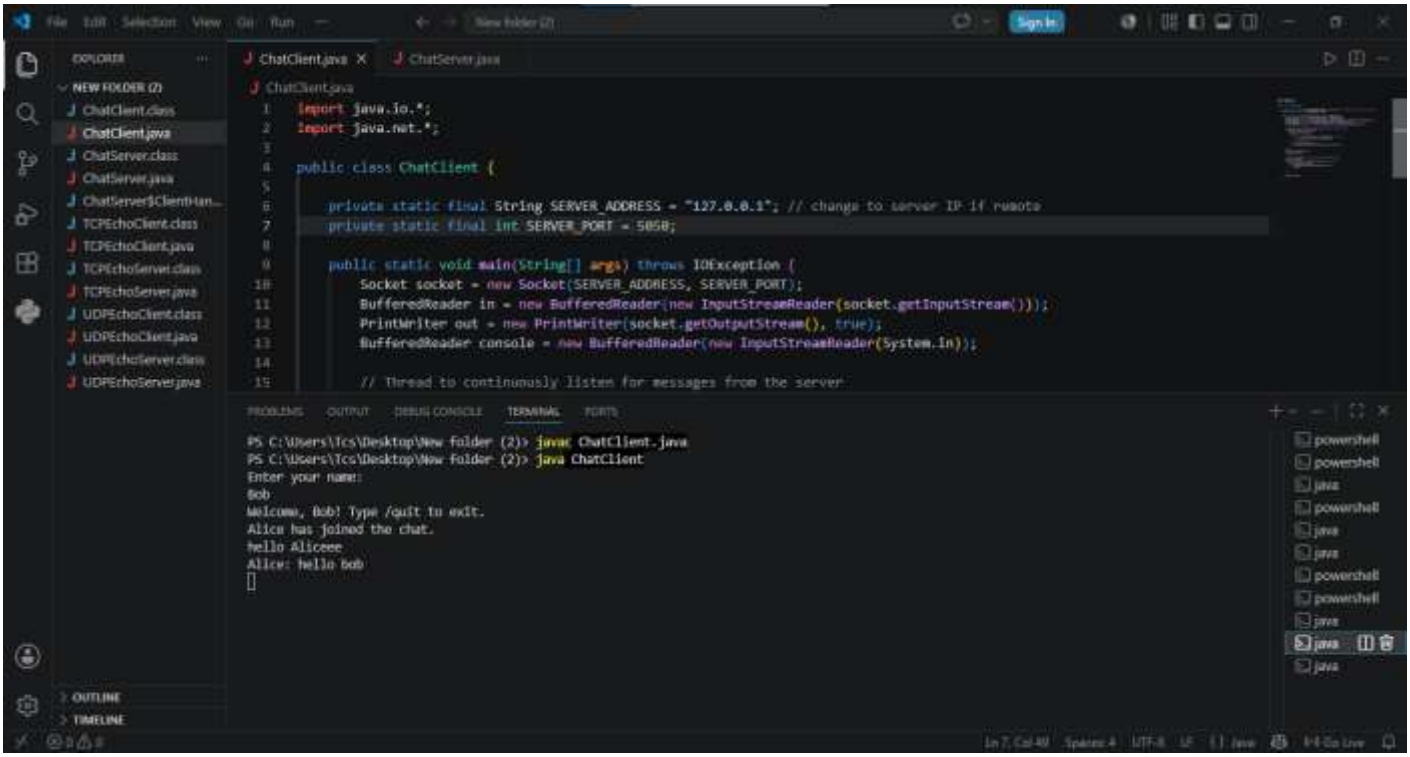

LINKSYS
Wireless
Wireless Security
Security Mode: WEP
Encryption: 40/64-Bit(10 Hex digit)
Passphrase:
Key1: 0123456789
PC0
Physical
Config
Desktop
IP Configuration
PC Wireless
Infrastructure Mode
You have successfully connected to the access point
WEP Key Needed for Connection

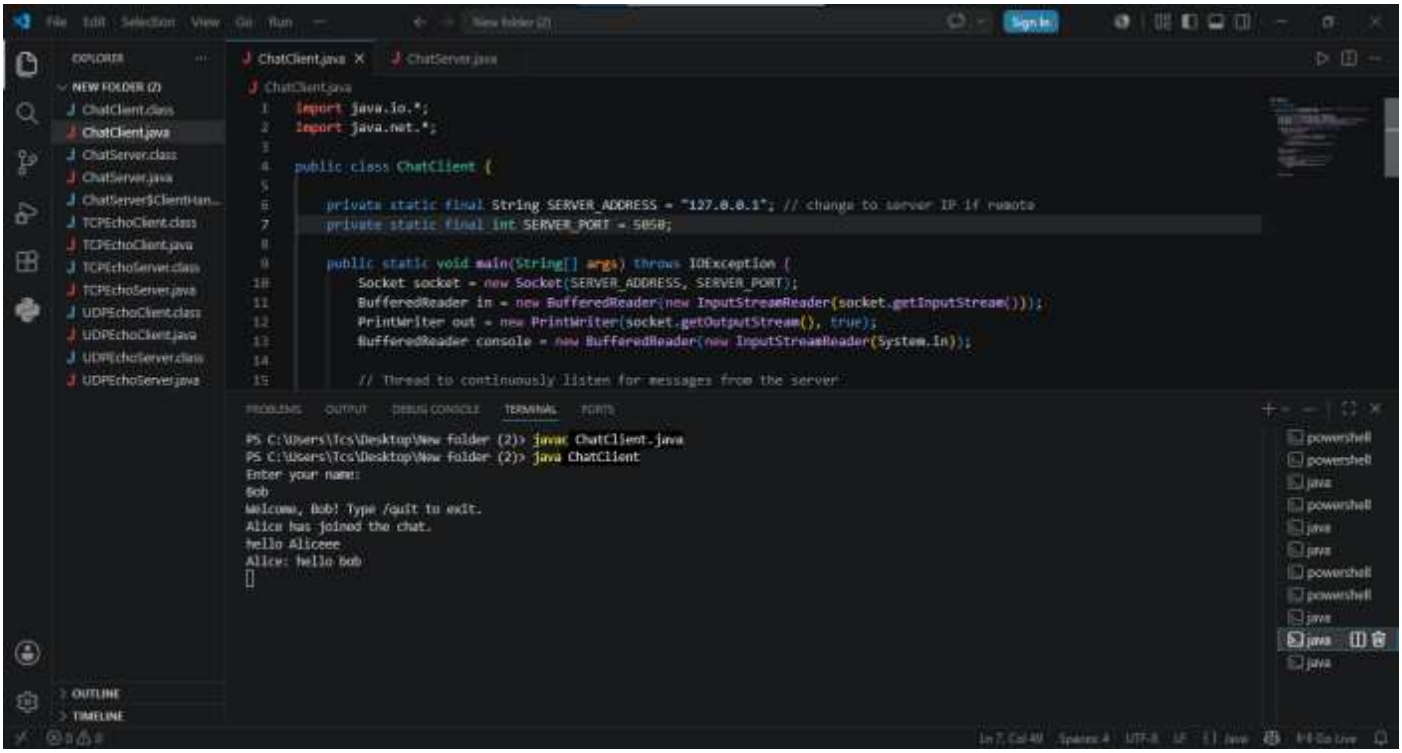

Cisco Packet Tracer - Logical network topology
File Edit Search View Tools Window Help
Logical
We also configured a separate static host route for the host 30.0.0.100/8...
Router0
Router1
PC0
PC1
PC2
PC2
Simulation

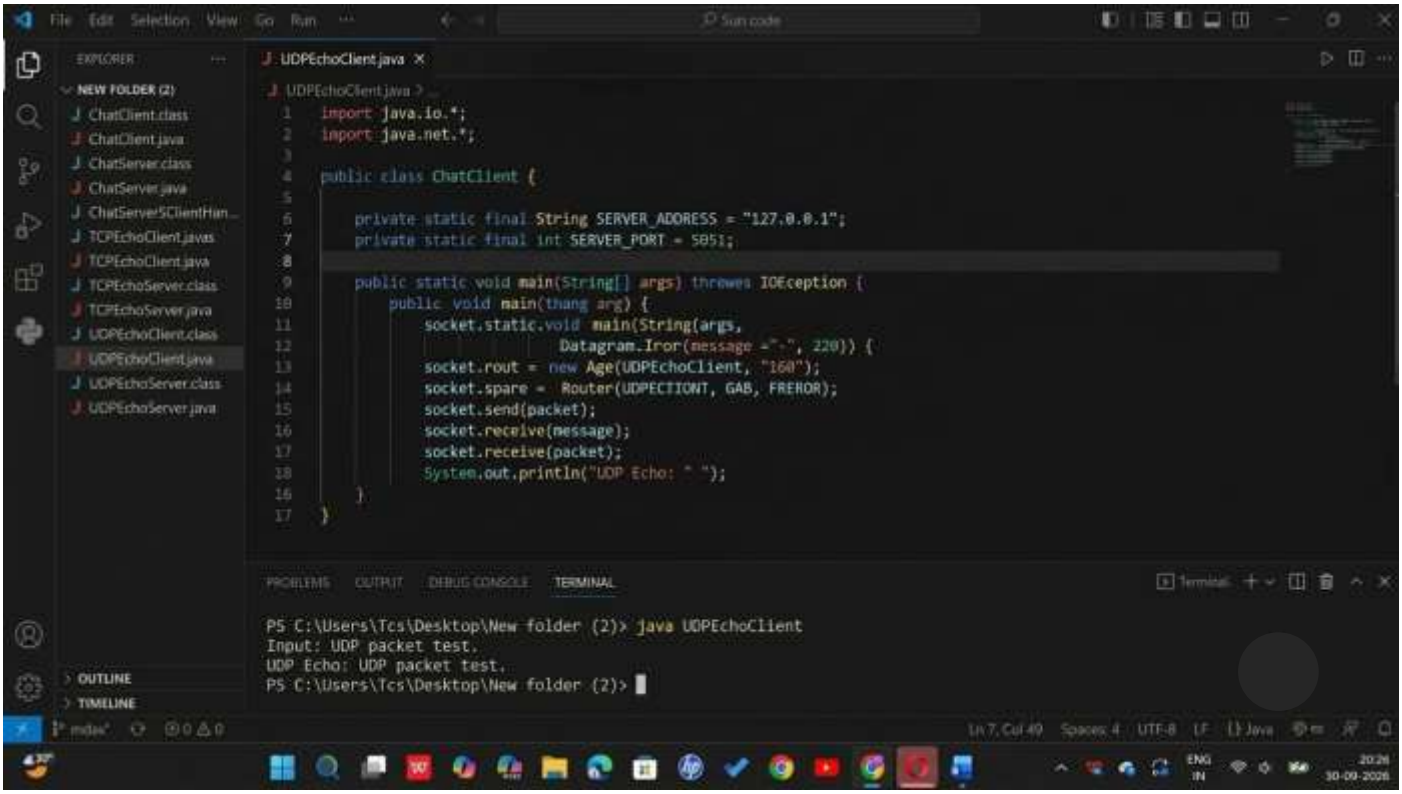

Cisco Packet Tracer - Logical network topology
File Edit Search View Tools Window Help
Logical
Gig0/0
Gig0/1
Gig0/0
Router0
Gig0/1
Gig0/1
Router1
Gig0/1
Server0
.100
.10
.20
.10
.20
PC0
PC1
PC2
PC3
Simulation

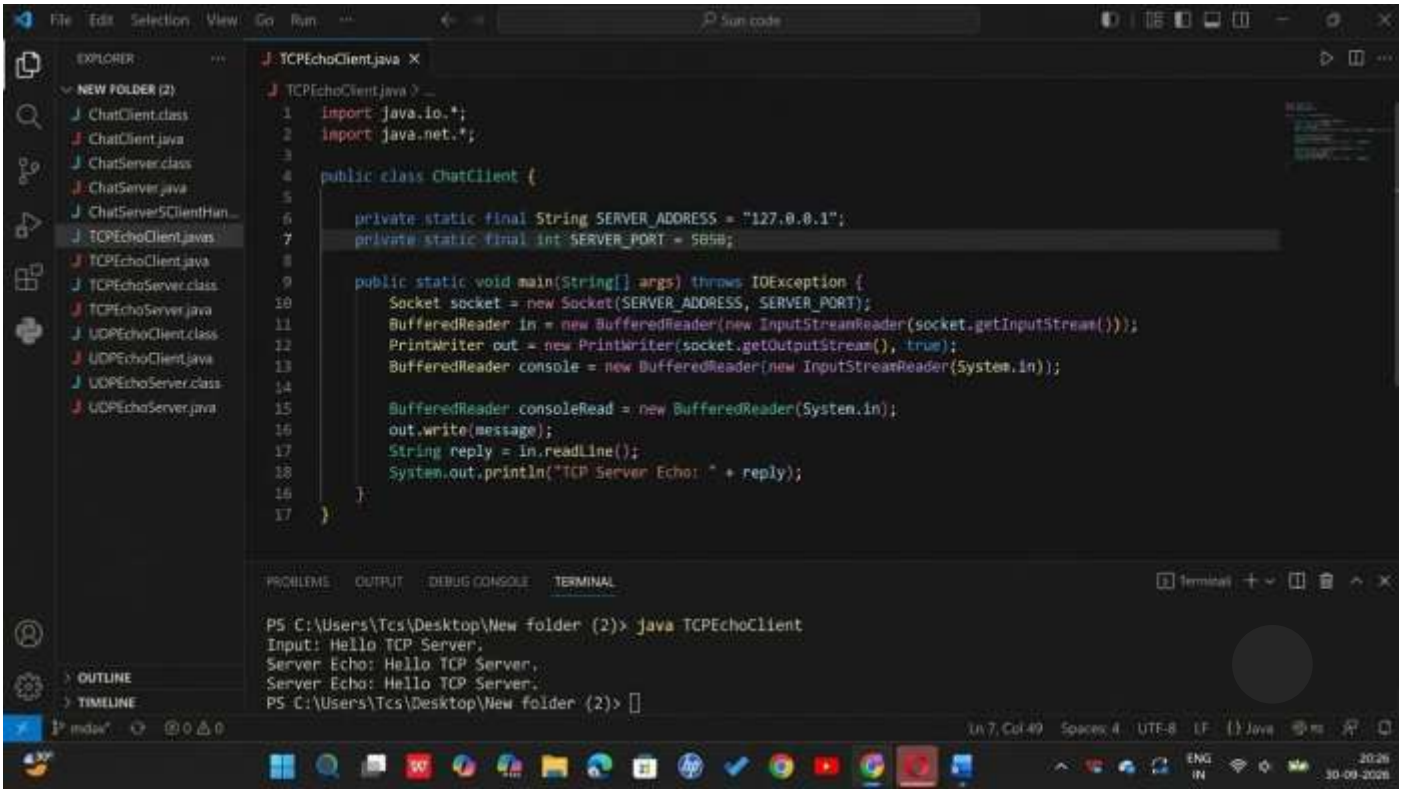

Cisco Packet Tracer - Logical network topology
File Edit Search View Tools Window Help
Logical
Router0
Router1
PC0
PC1
PC2
PC2


ChatClient.java
ChatServer.java
NEW FOLDER (2)
ChatClient.class
ChatClient.java
ChatServer.class
ChatServer.java
ChatServer$ClientHan...
TCPEchoClient.class
TCPEchoClient.java
TCPEchoServer.class
TCPEchoServer.java
UDPEchoClient.class
UDPEchoClient.java
UDPEchoServer.class
UDPEchoServer.java
import java.io.*;
import java.net.*;
public class ChatClient {
private static final String SERVER_ADDRESS = "127.0.0.1"; // change to server IP if remote
private static final int SERVER_PORT = 5050;
public static void main(String[] args) throws IOException {
Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
// Thread to continuously listen for messages from the server
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\Tcs\Desktop\New Folder (2)> java ChatClient
Enter your name:
Alice
Welcome, Alice! Type /quit to exit.
Bob: hello Aliceee
hello bob
OUTLINE
TIMELINE


ChatClient.java
ChatServer.java
NEW FOLDER (2)
ChatClient.class
ChatClient.java
ChatServer.class
ChatServer.java
ChatServer$ClientHan...
TCPEchoClient.class
TCPEchoClient.java
TCPEchoServer.class
TCPEchoServer.java
UDPEchoClient.class
UDPEchoClient.java
UDPEchoServer.class
UDPEchoServer.java
import java.io.*;
import java.net.*;
public class ChatClient {
private static final String SERVER_ADDRESS = "127.0.0.1"; // change to server IP if remote
private static final int SERVER_PORT = 5050;
public static void main(String[] args) throws IOException {
Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
// Thread to continuously listen for messages from the server
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\Tcs\Desktop\New Folder (2)> javac ChatClient.java
PS C:\Users\Tcs\Desktop\New Folder (2)> java ChatClient
Enter your name:
Bob
Welcome, Bob! Type /quit to exit.
Alice has joined the chat.
hello Aliceee
Alice: hello Bob
OUTLINE
TIMELINE


ChatClient.java
ChatServer.java
NEW FOLDER (2)
ChatClient.class
ChatClient.java
ChatServer.class
ChatServer.java
TCPEchoClient.class
TCPEchoClient.java
TCPEchoServer.class
TCPEchoServer.java
UDPEchoClient.class
UDPEchoClient.java
UDPEchoServer.class
UDPEchoServer.java
import java.io.*;
import java.net.*;
public class ChatClient {
private static final String SERVER_ADDRESS = "127.0.0.1"; // change to server IP if remote
private static final int SERVER_PORT = 5050;
public static void main(String[] args) throws IOException {
Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
// Thread to continuously listen for messages from the server
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\Tcs\Desktop\New folder (2)> javac ChatClient.java
PS C:\Users\Tcs\Desktop\New folder (2)> java ChatClient
Enter your name:
Bob
Welcome, Bob! Type /quit to exit.
Alice has joined the chat.
hello Aliceee
Alice: hello Bob
OUTLINE
TIMELINE


UDPEchoClient.java
NEW FOLDER (2)
ChatClient.class
ChatClient.java
ChatServer.class
ChatServer.java
TCPEchoClient.java
TCPEchoClient.java
TCPEchoServer.class
TCPEchoServer.java
UDPEchoClient.class
UDPEchoClient.java
UDPEchoServer.class
UDPEchoServer.java
import java.io.*;
import java.net.*;
public class ChatClient {
private static final String SERVER_ADDRESS = "127.0.0.1";
private static final int SERVER_PORT = 5051;
public static void main(String[] args) throwes IOEception {
public void main(thang arg) {
socket.static.void main(String(args,
Datagram.Iror(message ="-", 220)) {
socket.rout = new Age(UDPEchoClient, "160");
socket.spare = Router(UDPECTIONT, GAB, FREROR);
socket.send(packet);
socket.receive(message);
socket.receive(packet);
System.out.println("UDP Echo: " ");
}
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PS C:\Users\Tcs\Desktop\New folder (2)> java UDPEchoClient
Input: UDP packet test.
UDP Echo: UDP packet test.
PS C:\Users\Tcs\Desktop\New folder (2)>
OUTLINE
TIMELINE


File Edit Selection View Go Run
Search code
EXPLORER
NEW FOLDER (2)
ChatClient.class
ChatClient.java
ChatServer.class
ChatServer.java
ChatServerSClientHan...
TCPEchoClient.javas
TCPEchoClient.java
TCPEchoServer.class
TCPEchoServer.java
UDPEchoClient.class
UDPEchoClient.java
UDPEchoServer.class
UDPEchoServer.java
TCPEchoClient.java
1 import java.io.*;
2 import java.net.*;
3
4 public class ChatClient {
5
6 private static final String SERVER_ADDRESS = "127.0.0.1";
7 private static final int SERVER_PORT = 5858;
8
9 public static void main(String[] args) throws IOException {
10 Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
11 BufferedReader in = new BufferedReader(new InputStreamReader(socket.getInputStream()));
12 PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
13 BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
14
15 BufferedReader consoleRead = new BufferedReader(System.in);
16 out.write(message);
17 String reply = in.readLine();
18 System.out.println("TCP Server Echo: " + reply);
16 }
17 }
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Terminal
PS C:\Users\Tcs\Desktop\New folder (2)> java TCPEchoClient
Input: Hello TCP Server.
Server Echo: Hello TCP Server.
Server Echo: Hello TCP Server.
PS C:\Users\Tcs\Desktop\New folder (2)>
OUTLINE
TIMELINE
Ln 7, Col 49 Spaces: 4 UTF-8 LF Java
ENG
IN
20:26
30-09-2026